

Supplementary Materials: Mathematical Derivations and Lean 4 Formalizations

Overview

This supplementary document provides the formal mathematical framework underlying the core claims presented in the main manuscript. The theorems below have been formalized and verified using the Lean 4 interactive theorem prover. Formalization ensures that the transitions from foundational physical primitives (symmetries, topology, and thermodynamics) to the complex structural properties of consciousness (boundaries, unity, and reflexivity) are stated precisely rather than metaphorically. It does not by itself make them physically true. The development now declares *no* axioms — `#print axioms` reports only Lean’s three foundational ones — but the physical postulates have not vanished; they have moved into class fields that each instance must discharge, and into the hypotheses of individual theorems. Why a standalone axiom is the wrong shape for a physical postulate, and how such an axiom can prove `False` while leaving the build green, is the soundness subsection of the main text. Where the Lean statement is weaker than the informal theorem above it, we say so explicitly in the implementation note. The complete Lean 4 source code is publicly available at https://github.com/bobaseb/physical_primitives_of_mind, and web-accessible documentation is available at https://bobaseb.github.io/physical_primitives_of_mind/.

1 Symmetry Breaking and the Emergence of Boundaries

The framework’s first physical motif is that a perfectly symmetric universe cannot support agency. The formal result is narrower: under the stated vacuum and connectedness hypotheses, a field reaching two vacuum components must leave the vacuum somewhere.

Let $\mathcal{M}$ be a pseudo-Riemannian manifold representing spacetime, and let $\phi : \mathcal{M} \rightarrow \mathcal{V}$ be a physical field mapping to a vacuum manifold $\mathcal{V}$ (the space of energy minima). Let G be a continuous symmetry group acting on $\mathcal{V}$.

Theorem 1 (Topological Defects from Broken Symmetry): If the continuous symmetry G is spontaneously broken such that $\mathcal{V} \cong G/H$ (where H is the unbroken subgroup), and the fundamental group $\pi_1(\mathcal{V}) \neq 0$, then field configurations ϕ defined on a spatial boundary $S^1 \subset \mathcal{M}$ cannot necessarily be smoothly deformed to a constant vacuum state.

Lean 4 Implementation (Phase1_Primitives.lean): We formalize this by defining the set of degenerate energy minima `DynamicalVacuum` and proving `boundary_defect_forces_interior_vacuum_break`: if a boundary field configuration ϕ_∂ is not null-homotopic, then it admits no continuous extension over a contractible interior. This geometric obstruction forms the fundamental physical boundary separating an “inside” from an “outside.” Note the direction: this result establishes that a defect *obstructs extension*, not that a defect must exist. The companion lemma `contractible_interior_forces_trivial_boundary` (renamed from `defect_inevitability`, whose old name asserted the opposite of its statement) proves the contrapositive.

The direction that *does* produce a defect is now proved, at the level of π_0 . `exists_notMem_of_no_common_preconnected` states it in the form a witness can discharge — if no preconnected subset of the vacuum manifold M contains both $\phi(x_1)$ and $\phi(x_2)$, and the substrate is connected, then ϕ leaves M somewhere — and `exists_notMem_of_connectedComponentIn` restates it with $\phi(x_2) \notin$ the connected component of M containing $\phi(x_1)$. The proof uses the connectedness of continuous images and `IsPreconnected.subset_connectedComponentIn`, and nothing else; no hypothesis mentions energy, a potential or a symmetry. This is the domain-wall case of Kibble’s classification and we take only that case: $\pi_1 \neq 0$ forcing vortices and $\pi_2 \neq 0$ forcing monopoles are recorded as available and not taken, since the manuscript’s argument uses domain walls alone. Theorem 1 as stated above, in terms

of π_1 and a coset space G/H , therefore remains *stronger than what is formalized*; what is formalized is its π_0 analogue, and the gap between them is the higher homotopy, not the mechanism. Separately, `spontaneous_symmetry_breaking` shows a global energy minimiser lies in the vacuum manifold almost everywhere; this is now axiom-free, the pointwise step being `pointwise_vacuum_of_global_min`. `Examples.lean` §11 witnesses that result on the double well $V(v) = (v^2 - 1)^2$ over a two-point spacetime carrying a Dirac probability measure. Two features of the witness are deliberate. Its minimisation hypothesis is *proved* rather than assumed — everywhere else in the development the existence of a global minimiser is a hypothesis, since no compactness or direct-method argument is formalized. And the almost-everywhere in the conclusion is shown necessary: the minimising field sits at the vacuum on the support and at 0 off it, so the pointwise version of the statement is false of the very field the theorem applies to. The same section instantiates the $\mathbb{Z}_2$ action $v \mapsto \pm v$ and discharges the energy invariance, giving the first instance of `SymmetryInvariantAction`; the vacuum manifold $\{-1, 1\}$ is proved degenerate and the two minima exchanged by the symmetry, so a minimiser and its image have equal energy and differ. `Examples.lean` §11.1 then supplies the witness for the π_0 theorem on the same well. The vacuum manifold is *computed* there, `wellVacuum.eq` proving `DynamicalVacuum` $V = \{-1, 1\}$ exactly rather than merely containing both minima — which matters, because a larger vacuum set might have been connected — and `wellVacuum.separated` proves the disconnection in the form the theorem consumes, by the observation that a preconnected subset of $\mathbb{R}$ spanning -1 and 1 contains 0 , which is not a minimum. `wellV.domain.wall` is the resulting statement about *every* continuous field on *any* connected substrate reaching both minima. Non-emptiness of that class is checked separately rather than assumed: the clipped identity on $\mathbb{R}$ satisfies the hypotheses, and `kink_zero_notMem` locates its forced wall at the origin. No conserved quantity is constructed from any of this, and there is still no Noether theorem.

2 The Thermodynamic Constraint: Landauer’s Principle

Once a boundary is established, it is subjected to external perturbations. Because the internal phase space of any localized physical system is finite, the system must overwrite internal states to adapt. That premise is discharged by theorem rather than by assertion in `Phase1.PhaseSpaceCapacity.lean`: a distribution on a finite state space X carries at most $\log |X|$ nats, with equality only at the uniform law (`shannon_entropy.eq_log_card_iff`), and a perturbation stream of length k over an alphabet P with $|P|^k > |X|$ therefore cannot be recorded injectively (`absorb_not_injective`), whatever the update map is and even when every individual step of it is reversible.

Let the global phase space $\Omega = \Omega_{sys} \times \Omega_{bath}$ undergo globally reversible Hamiltonian dynamics, modeled as an injective mapping $f : \Omega \rightarrow \Omega$.

Theorem 2 (Thermodynamic Erasure): If the local transition map on the system $\pi_{sys} \circ f$ is not injective (i.e., multiple initial states map to the same final state, contracting the local phase space), then by the conservation of global phase space volume (Liouville’s theorem), the bath phase space must expand, resulting in an irreversible transfer of heat to the environment: $\Delta Q \geq k_B T \ln 2$.

The same heat, in three vocabularies (Phase3.LandauerBridge.lean). Derivation 2 charges a register for destroying information and Derivation 3 charges a structure for keeping information it cannot use. `PredictiveDissipation.ofLandauer` builds the predictive structure of a given register: its thermal scale is that register’s temperature, its dissipated work is that register’s Landauer heat, and Still’s bound is derived from `landauer_bound` rather than assumed a second time. The single hypothesis is the identification $(I_{mem} - I_{pred}) \leq S(id) - S(t)$ — the memory wasted is no larger than the entropy erased — and it is a restriction, not bookkeeping: on an injective update the right-hand side is zero and the inequality forces the waste to vanish (`nonpredictive.eq_zero_of_injective`), which is the Norton–Shenker distinction in informational form. The measure-theoretic statement of Landauer (`Phase3.MeasureThermodynamics`) is specialised in the same file rather than left standing beside the combinatorial one: at counting measure on a finite register, `continuous_entropy` is `boltzmann_entropy`, volume contraction is non-injectivity, and `landauer_heat_bound` is T times the entropy erased (`landauer_heat_bound_count.eq`). The witness is the one-bit eraser, where a register destroying $\log 2$ pays for a memory wasting $\log 2$, with equality (`Examples` §18.5).

Lean 4 Implementation (Phase3.CombinatorialThermodynamics.lean): We explicitly decouple logical erasure from physical erasure. Lean verifies that any contraction in the *physical* state space ($\neg \text{Function.Injective}(t_{sys})$) under globally reversible dynamics mathematically necessitates heat dissipation. The combinatorial core, `landauer_from_reversibility` and `second_law`, is fully derived. Finiteness of the phase space is what converts the physically checkable hypothesis into the one the bound needs: on a finite state space a map that fails to reach every state is not injective (`is_eraser_`

of `not_surjective`), so an unreachable state suffices for $\Delta Q > 0$ (`finite_phase_space_dissipates`). The hypothesis is fenced: `Nat.succ` misses 0 and is injective, so the implication is false on an infinite phase space, and `Examples.lean` §19 records this beside the witnesses. The Norton–Shenker distinction survives the identification — retaining the incoming perturbation is injective and free (`keepRecord_injective`), refreshing the register that held it is not (`absorbStep_is_erasure`). Converting it into a statement about heat requires the postulate $\Delta Q = T\Delta S_{\text{bath}}$, which is carried as the class field `StatisticalMechanics.heat_eq`; `Examples.lean` exhibits a one-bit erasure system with an explicit bath that discharges it, so the postulate is demonstrably satisfiable. The boundary is thus forced to operate as a dissipative structure.

3 Structural Resonance and Least Action

To avoid catastrophic dissipation, the dissipative structure evolves to minimize entropy production. Let $P(\text{ext})$ be the probability measure of external environmental perturbations, and $Q(\text{int})$ be the internal transition probabilities of the system.

The total entropy production σ is bounded from below by the Kullback-Leibler divergence:

$$\sigma \geq \frac{1}{\Delta t} D_{KL}(P(\text{ext}) || Q(\text{int})) \quad (1)$$

What the bound above does not support. One natural reading of it runs: by the Principle of Least Action the system undergoes gradient descent on σ , forcing $\lim_{t \rightarrow \infty} D_{KL}(P || Q) = 0$. That conclusion does not follow. Rearranged it says $D_{KL} \leq \Delta t \sigma$, so descending on σ lowers a *bound* on the divergence; but a driven system settles into a non-equilibrium steady state with $\sigma_{\text{NESS}} > 0$, and the bound then permits any divergence up to $\Delta t \sigma_{\text{NESS}}$. The limit is not merely unproved here. It does not follow.

Theorem 3 (Nonpredictive information bounds dissipation): Let X_t be the system’s state and S_t an environmental signal evolving by a dynamics of its own on which the system does not act. Then

$$I(X_t; S_{t+1}) \leq I(X_t; S_t), \quad W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})], \quad (2)$$

the second being the bound of Still, Sivak, Bell and Crooks (*Phys. Rev. Lett.* **109**, 120604, 2012). Consequently $W_{\text{diss}} \geq 0$, and any bound on the dissipated work is a bound on the nonpredictive information $I(X_t; S_t) - I(X_t; S_{t+1})$.

The first inequality is the data processing inequality for the Markov chain $X_t \rightarrow S_t \rightarrow S_{t+1}$, proved in `Phase3.PredictiveThermodynamics.lean` as `predictiveInfo_le_mutualInfo`; mutual information is defined there from Mathlib’s `klDiv` as the divergence of the joint law from the product of its marginals, since Mathlib carries no mutual information of its own. The second inequality is the class field `PredictiveDissipation.still_bound`, an irreducible physical postulate for the same reason as every other one here — its axiom form is refutable, and `Examples.lean` §18.4 refutes it. Everything downstream is derived: non-negativity of the dissipated work, the bound of the nonpredictive information by the dissipation budget, and the statement that a quasi-static drive leaves every retained bit predictive. `Examples.lean` §18 inhabits the class twice over one correlated two-bit law — a frozen signal, permitted to dissipate nothing, and a scrambled one, forced to dissipate its entire memory — and fences the theorem with a Markov kernel on the pair that writes the system’s state into the signal and thereby *raises* mutual information, so the no-back-action hypothesis is shown to carry weight rather than to decorate. The equality of Kawai, Parrondo and Van den Broeck (*Phys. Rev. Lett.* **98**, 080602, 2007), $\langle W_{\text{diss}} \rangle = k_B T D_{KL}(\text{forward} || \text{reverse})$, is carried as a predicate on an instance and yields `nonpredictive_le_arrow`: the irreversibility of the dynamics is at least the memory that fails to earn its keep.

This is what survives of the claim that “prediction error minimization” is a physical thermodynamic attractor: retained information that does not predict is paid for in heat. It is not a derivation of the Free Energy Principle and it does not ground symbols. The information-theoretic KL bound is formalized in `Phase3.KLBound.lean` as the class field `StructuralResonance.kl_bound` — an irreducible physical postulate attached to a specific system rather than asserted globally, the earlier `kl_bound_axiom` formulation having been inconsistent. Gibbs’ inequality (KL non-negativity) is proved as a genuine theorem from $\log x \leq x - 1$; `structural_resonance_bound` restates the axiom in the form $D_{KL} \leq \Delta t \cdot \sigma$, and `discrete_entropy_rate_nonneg` combines the two to derive $\sigma \geq 0$. The gradient-flow half of that reading is formalized against the entropy production functional itself: `Phase8.ContinuousField.lean`

§7 derives the Fréchet derivative of σ with respect to the coupling kernel, identifies the differentiated functional with `entropy_production_rate` on finite substrates carrying counting measure, and concludes in `structural_resonance_decreases_entropy_production` that σ is non-increasing along the resulting flow; `Examples.lean` §5 exhibits a two-site substrate on which σ decays as e^{-2t} . The limit $\lim_{t \rightarrow \infty} D_{KL} = 0$ is not among the consequences and is claimed nowhere in either document: monotone descent plus the bound $D_{KL} \leq \Delta t \sigma$ cannot force it, because the descent terminates at a positive steady-state value of σ . It is tempting to file this as a missing estimate — a Łojasiewicz or coercivity argument yet to be supplied — and that would misdescribe it: the gap is in the inference and not in the analysis, so it is closed by replacing the inference, as the theorem above does. The link is also restricted to finite substrates and to a fixed phase field, though the first restriction is narrower than the shape of σ suggests. `Phase8.ContinuousField.lean` §8 is where that is made precise: σ is a quadratic functional of an affine image of the kernel, so `hasFDerivAt_quadratic_of_affine` differentiates it by the chain rule with no limiting argument, and `hasFDerivAt_sigmaOfKernel_of_operator` recovers §7’s gradient from that abstract lemma. The continuum case was consequently blocked on exhibiting the drift map as a bounded operator between L^2 spaces — a Cauchy–Schwarz estimate — rather than on differentiation under the integral sign, and that operator is now built (`kernelCLM`, `continuumDriftCLM`, with operator norm at most $\mu(M)^{1/2}$). One further step was needed that the recorded scope had missed: the descent theorems want the derivative as a gradient *vector*, which needs the adjoint, available because L^2 is complete. `hasFDerivAt_quadratic_grad` supplies $\nabla\sigma(K) = (2/D)A^*(AK + \omega)$ for an arbitrary bounded operator between real Hilbert spaces, so `structural_resonance_decreases_sigmaContinuum` is the continuum descent result with no finiteness hypothesis anywhere in it, and `Examples.lean` §20 runs it on Lebesgue measure restricted to $(0, 1)$. A separate result in the same file concerns the target of the descent rather than the descent itself, and it is now unconditional on the class of kernels the framework uses. `phase_locked_achieves_minimum_entropy` proved that the phase-locked field minimizes σ only among competitors sharing its spatial mean drift, which made the content Jensen’s inequality alone; the note recorded that removing the restriction would mean modelling how the mean drift varies with the competitor. It does not. For a symmetric kernel the integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric in its two sites, so by Fubini its double integral equals its own negative and vanishes (`coupling_integral_eq_zero`); the total drift is consequently the same for every phase field (`total_drift_eq_of_symm`), the phase-locked configuration fixes its value, and the mean-matching hypothesis is a theorem (`mean_drift_of_symm`) rather than an assumption. `phase_locked_minimizes_entropy_of_symm` is the resulting statement, quantified over all competitor fields. Symmetry is imposed as a hypothesis rather than a field of `ContinuousNeuralField`, since the results quantify over instances, and it is load-bearing: `Examples.lean` §16 builds a one-way coupling on the two-site substrate whose phase-locked field is strictly beaten (`asymSys_locked_not_minimal`, $\sigma = 1/2$ against 1), with `asymSys_mean_drift_fails` isolating the mechanism — there the total drift is not conserved, so the variance bound has nothing to stand on. On the witness side, `Examples.lean` §2 now carries two instances rather than one. The original is perfectly resonant, $D_{KL} = 0$, and so discharges the postulate trivially. The second (`boolDetuned`) is the one-bit eraser of §1 driven by a point-mass environment against uniform internal statistics: there $D_{KL} = \log 2 > 0$ and $\sigma = \log 2 > 0$, and `boolDetuned_tight` proves the postulate holds with *equality*. The bound is therefore attained and cannot be strengthened to a strict inequality.

4 Macroscopic Scaling and Kuramoto Phase Synchronization

As the system scales, it forms a continuous macroscopic field. We model the local phases θ_i as coupled non-linear oscillators via the Kuramoto model.

Scope of that sentence. `mesh_refinement_convergence` proves convergence of one scalar coupling energy. It produces neither a continuum kernel nor field dynamics: `edge_region` has no $M \times M$ product structure from which such a kernel could be recovered, and the vertex-supported candidate is invisible to the continuum functional on an atomless substrate. The later symmetric kernel is therefore a modelling input. Its identification with the brain’s endogenous *electromagnetic* field is a further empirical commitment about cortex, argued in the main text from the ephaptic-coupling literature and stated with four falsification conditions. Nothing in this section mathematically establishes either step.

For a system with coupling strength K and thermodynamic noise D , the dynamic evolution is:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) + \eta_i(t) \quad (3)$$

Theorem 4 (Critical Phase Transition): When the integrated spatial coupling K exceeds the thermodynamic noise threshold $K_c = 2D$, the oscillators undergo a sudden phase transition into a synchronized, phase-locked state ($r > 0$).

Lean 4 Implementation (Phase4.KuramotoDynamics.lean): We formalize the Lyapunov potential $V(\theta) = -\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_j - \theta_i)$ of the coupled field and prove two directions: `phase_locked_minimizes_potential` (the phase-locked configuration attains the global minimum whenever all couplings are positive) and `potential_min_implies_phase_locked` (any minimizer is phase-locked). Note the scope: this is a statement about the *static* potential landscape and holds for all positive coupling matrices. The critical threshold $K_c = 2D$ does not appear in this file. It is formalized separately, in `Phase8.SelfConsistency.lean` (see below), as a statement about the mean-field self-consistency equation with the stationary density assumed; deriving that density from the noise-driven stochastic dynamics would require the spectral theory of the associated Fokker-Planck operator, which is not available in Mathlib. The threshold is also validated numerically (`simulations/kuramoto.py`).

Positivity, and the regime these theorems cover. The hypothesis “all couplings positive” is not a technical convenience and it is not free. `KuramotoSystem` carries $A : V \rightarrow V \rightarrow \mathbb{R}$, symmetric and otherwise unconstrained, so signed — frustrated — couplings are representable; but every result in this development that identifies *where* the dynamics ends assumes positivity. `phase_locked_minimizes_potential` and `potential_min_iff_phase_locked` assume $A_{ij} > 0$; the class field `ThermodynamicCover.A.pos` demands it of every instance, and so of every conclusion of Theorem 5; and `kuramoto_tendsto_global_minimum` requires the strictly stronger uniform bound $0 < a \leq A_{ij}$, since a enters the Łojasiewicz constant. Such a network is an unfrustrated ferromagnet with one minimum up to a global phase shift — exactly not the near-degenerate landscape a memory argument from geometric frustration requires. By contrast, nothing about the sign of A is used by `is_kuramoto_trajectory_exists` and its uniqueness counterpart, by the Lyapunov identity `dV_dt_le_zero` on which the whole descent argument rests, by `dynamic_potential_antitone` and `dynamic_potential_tendsto`, or by `velocity_sq_tendsto_zero`. Since positivity is a *field* of `ThermodynamicCover` rather than a hypothesis distributed across the development, it enters the chain at one node and not before it. So *that* the motion stops holds for frustrated systems as well; *where* it stops does not. On the continuum side `phase_locked_minimizes_entropy_of_symm` assumes only symmetry and so admits signed kernels, but it takes the existence of a dynamically phase-locked field as a hypothesis, which is precisely what frustration threatens. The main text scopes the frustration claim out of the chain accordingly, rather than repeating it beside theorems that exclude it.

Twisted gluing does not recover content from phase frustration (Phase5.TwistedGluing.lean). We also tested the natural weakening of exact overlap agreement: local states may differ by a phase on each overlap. `gluesUpToPhase_of_isCoboundary` proves that coboundary offsets glue after one rotation per patch, while `isCoboundary_of_gluesUpToPhase` proves the converse when the phase action is free on overlaps. Any Kuramoto configuration, including a twisted or splay state, supplies absolute patch phases ϕ_i , so its offsets $\phi_i - \phi_j$ are automatically a coboundary (`isCoboundary_of_phaseField`) and the twisted family glues (`gluesUpToPhase_of_phaseField`). The formal `PhaseObstruction` quotient is deliberately not called Čech cohomology because its total cochains still assign irrelevant values to patch pairs with empty overlap.

The data give a narrower positive answer. Townsend et al. (2015) identify phase singularities in multi-electrode primate LFP recordings by an integer winding number around electrode loops, and the associated complex waves modulate spiking; Xu et al. (2023) find task-dependent spiral centres and rotation directions in human fMRI phase maps. Around such a zero-amplitude point the circle-valued phase need not admit one real-valued lift. Local lifts differ on overlaps by integer multiples of 2π , supplying measured transition data without an arbitrary gauge field. `Phase5.PhaseLifts.lean` formalizes the finite electrode loop: `loopWinding` sums its integer transitions, `loopWinding.eq_zero_of_hasGlobalLift` proves that transitions from one global lift telescope to zero, and `threeElectrodeWinding.not_hasGlobalLift` gives a winding-one witness. This does not rescue `PhaseObstruction`: that quotient uses circle-valued differences of absolute patch phases, while the measured obstruction uses integer transitions between local lifts.

The cover-level specification has cortical surface M , analytic signal $z = Ae^{i\phi}$, singularity set Z_t , punctured domain $M_t^* = M \setminus Z_t$, and measurement map $q_t = z/|z| : M_t^* \rightarrow S^1$. A trial-independent electrode or source-space cover of M_t^* supplies local real lifts; their differences divided by 2π are integer transitions on connected overlaps, and nerve loops are the inputs to `loopWinding`. In finite-resolution data the amplitude threshold defining Z_t , the loop geometry and the required cross-scale persistence are fixed using training or baseline data, never the content labels of test trials.

The proposed content-sensitive test compares nested preregistered classifiers. The control model

contains stimulus, task, behavioural response, eye movement and arousal; the augmented model adds signed winding, source-localized singularity position, rotation direction and lifetime. Evaluation uses held-out participants and sessions, with label permutations within task, stimulus and arousal strata. The content hypothesis requires replicated incremental out-of-sample discrimination, including under a changed or absent motor report. This controls the known task dependence rather than mistaking it for content. A positive result would be an empirical association, not an identification of a topological sector with phenomenal content; a null increment rejects this proposed role. The construction and test remain outside the conditional chain, and a full nerve-indexed integer complex is deferred until the measurement protocol warrants it.

Lean 4 Implementation (Phase8ContinuousField.lean): What the continuous-field file does carry is the threshold value itself, `critical_coupling D = 2 * D`, and a predicate `exhibits_phase_transition` for a substrate lying above it. Two cautions. First, the predicate is not connected to `is_continuous_kuramoto_trajectory`, `entropy_production_rate`, or the order parameter of `Phase4`. What it is connected to is the mean-field self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$, whose bifurcation at $K = 2D$ we formalize in both directions (see the next paragraph): `exhibits_phase_transition_coherent` shows that a substrate satisfying the predicate admits a solution with $0 < r \leq 1$, and `exhibits_phase_transition_unique_coherent` that it admits exactly one. So the threshold is not a bare stipulation, but what is proved is a statement about a stationary density rather than about a trajectory. Second, the predicate requires the substrate's `volume` to be a probability measure, and this hypothesis is load-bearing rather than decorative. It compares `mean_field_coupling`, the kernel averaged over both arguments, against $2D$; on a substrate of total mass m that average carries a factor m^2 , so without normalization the comparison tracks the substrate's size rather than its coupling and can be satisfied by any kernel at all. `Examples.lean §9` exhibits a system a factor of two below threshold whose unnormalized double integral nonetheless clears $2D$. Under the hypothesis, `mean_field_coupling_const` gives that a constant kernel has strength exactly K and `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$, witnessed in §9 on both sides of the threshold.

Lean 4 Implementation (Phase8SelfConsistency.lean): The self-consistency equation behind $K_c = 2D$ is formalized as real analysis. With the von Mises stationary density $\rho(\theta) \propto e^{a \cos \theta}$, $a = Kr/D$, taken as given (the algebraic step from the stationary Fokker–Planck equation to this density is verified independently at `simulations/fokker_planck_von_mises.py` in the repository), `besselRatio` is $I_1(a)/I_0(a)$ written as a ratio of interval integrals, Mathlib having no modified Bessel functions. `vonMisesM_eq_mul_vonMisesS` is the integration-by-parts identity $\int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} = a \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta}$, exact because $\sin(\pm\pi) = 0$ kills the boundary term. `vonMisesC2_nonneg` proves $I_2(a) \geq 0$ by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ through the reflections $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi/2 - \theta$, where the integrand becomes a product of two manifestly non-negative factors; `vonMisesS_le_half_vonMisesZ` restates it as $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$ and `besselRatio_le_half_self` as $R(a) \leq a/2$. That factor of $1/2$ is exactly what separates a threshold at D from one at $2D$; `subcritical_fixed_point_eq_zero` is the crude form obtained from $\sin^2 \leq 1$ and is kept to make the point visible. The subcritical theorem is `subcritical_fixed_point_eq_zero'`: for $0 \leq K < \text{critical_coupling } D$ and $D > 0$, the only $r \geq 0$ with $r = R(K, r)$ is $r = 0$. The supercritical theorem is `supercritical_fixed_point_exists`: for $K > \text{critical_coupling } D$ there is an r with $0 < r \leq 1$ and $r = R(K, r)$. Its proof needs no series expansion. Factoring $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the factorization the integration by parts supplies — it suffices that E is continuous with $E(0) = 1/2$: since $D/K < 1/2$, the map overshoots for small $r > 0$, while `besselRatio_le_one` (R is a mean of $\cos \theta$) makes it undershoot at $r = 1$, and `intermediate_value_Icc'` supplies the crossing. Continuity of E is Mathlib's continuity of a parametric interval integral, applied to the jointly continuous integrand. `critical_coupling_is_threshold` states the two halves together, and §9 of the file discharges the hypotheses of each at concrete values ($K = 3$, $D = 1$ above threshold; $K = 3$, $D = 2$ below it), so neither is vacuous. Section 8 of the file identifies the equation with the physics it is named for. `vonMisesDensity` is the normalized weight, proved a probability density on $[-\pi, \pi]$ (`vonMisesDensity_integral_eq_one`); `circularOrderParameter` is the average of $e^{i\theta}$ against a density, the continuum analogue of `order_parameter_complex` of `Phase4`; `circularOrderParameter_vonMises` proves that average equals $R(a)$, real because `vonMises_mean_sin` gives $\mathbb{E}_a[\sin \theta] = 0$; and `fixedPoint_iff_selfReproducing` states the equivalence: r solves $r = R(K, r)$ exactly when the density it induces has order parameter r . `incoherent_density_uniform` identifies the solution $r = 0$ with the uniform density, and `supercritical_coherent_density` restates the existence theorem as the existence of a stationary density with positive order parameter. Three results added since sharpen the threshold. `vonMisesC2_pos` upgrades the fold argument to a strict inequality for $a > 0$ — `foldB` is continuous, non-negative on $[0, \pi/4]$ and strictly positive on its interior, so

`intervalIntegral_pos_of_pos_on` applies — giving $\mathbb{E}_a[\sin^2 \theta] < 1/2$ and $R(a) < a/2$ for every $a > 0$. That closes the threshold case: `fixed_point_eq_zero_of_le_critical` covers all $K \leq 2D$, so the incoherent and coherent regimes now exhaust $K \geq 0$ with no gap, and $K = 2D$ falls on the incoherent side. `coherent_iff_sRatio_eq` reformulates the coherent branch: for $r \neq 0$, solving $r = R(K, r)$ is equivalent to $E(Kr/D) = D/K$, so the branch is a level set of E . `coherent_branch_continuous_at_threshold` then proves that for every $\varepsilon > 0$ there is a $\delta > 0$ such that on $(2D, 2D + \delta)$ every coherent solution satisfies $r < \varepsilon$ — the branch emerges continuously from zero, excluding a discontinuous jump. Its proof needs no monotonicity of E : on a compact interval bounded away from the origin the extreme value theorem gives a maximum strictly below $1/2$, and beyond that the crude bound $E(a) \leq 1/a$ suffices. Section 7 of the file closes uniqueness, which the paragraph above reduced to the injectivity of E on $(0, \infty)$, and closes it in the stronger form `vonMisesSRatio_strictAntiOn`: E is strictly decreasing on $[0, \infty)$. The expected route — differentiate the exponential family, obtain $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, and show that covariance positive — is not the route taken, and it was the harder one: it needs differentiation under the integral sign, and the covariance is not sign-definite pointwise, since the symmetrised form $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$ with $X = \cos \theta$ has an integrand changing sign with $X + X'$. Instead the target is kept as the two-point inequality $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: `vonMisesZ_eq_foldW` and `vonMisesC2_eq_foldA` carry both integrals onto $[0, \pi/2]$ by the two reflections already used for $I_2 \geq 0$, replacing $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$; there `crossIntegrand_nonneg` shows $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both being governed by $\cos x$ against $\cos y$. This is the same $X \mapsto -X$ symmetrisation the covariance form needed, performed once at the start rather than inside a double integral, and its analytic content is one identity: `cosh_mul_cosh_cross`, i.e. $\cosh(ap) \cosh(bq) \leq \cosh(bp) \cosh(aq)$ for $0 \leq a \leq b$, $0 \leq q \leq p$, from $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$ and $\cosh$ increasing in $|\cdot|$. *Then cross*: rather than integrating that inequality over a square, `foldA_mul_foldW_lt` takes α to be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a point where $\cos 2x_0 = \alpha$, supplied by `intermediate_value_lcc'`; evaluating at $y = x_0$ and integrating in x alone gives the conclusion, the $W(b, x_0)$ term vanishing by the choice of α . Strictness comes from `crossIntegrand_pos` on a subinterval missing x_0 . The same crossing argument run with $\cos \theta$ in place of $\sin^2 \theta$ needs no fold, since $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the tilting variable itself, and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$. The consequences are `coherent_fixed_point_unique` (at most one positive solution), `supercritical_fixed_point_existsUnique` and `supercritical_solution_set` (the non-negative solutions are exactly $\{0\}$ at or below threshold and $\{0, r\}$ above it), and `coherent_branch_strictMono`, which uses both halves: E decreasing puts the concentrations of two couplings in the order $a_1 < a_2$ through $E(a_i) = D/K_i$, and R increasing carries that order to $r_1 < r_2$. Scope: the density is assumed, not derived from the Fokker–Planck equation; nothing in this file differentiates anything, so E and R are reached as strictly monotone rather than as smooth, and the rate at which the branch leaves zero is the business of `Phase8.CriticalExponent.lean` below; dynamical selection is not established, and could not be here, since no theorem in the file mentions a trajectory; and the order parameter reached is that of a density, not of a trajectory — no theorem relates `circularOrderParameter` to `order_parameter_complex`, which would be the mean-field limit. Every declaration in the file depends only on `propext`, `Classical.choice` and `Quot.sound`.

Lean 4 Implementation (Phase8.CriticalExponent.lean): The rate is what separates the bifurcation from a generic saturating gate, and it costs the differentiation the file above avoids. `vmMoment n a` is the cosine moment $\int_{-\pi}^{\pi} \cos^n \theta e^{a \cos \theta}$, and `hasDerivAt_vmMoment` differentiates it in a under the integral sign, raising the cosine power by one; the interchange is Mathlib's dominated-convergence form for interval integrals, dominated by the constant $e^{|a|+1}$ on the unit ball around the parameter, which is where the compactness of $[-\pi, \pi]$ and the boundedness of $\cos$ do their work. Iterating gives `contDiff_two_vmMoment`, and `vonMisesSRatio_eq_moments` rewrites $E(a) = 1 - \text{vmMoment } 2a / \text{vmMoment } 0a$, so E inherits two derivatives from a quotient whose denominator is positive. The moments at $a = 0$ are elementary integrals of powers of $\cos$: $2\pi, 0, \pi, 0$ and $3\pi/4$. They give $E(0) = 1/2$, $E'(0) = 0$ — the first derivative vanishes by the symmetry that kills the odd moments — and $E''(0) = -1/8$, and Taylor's theorem in Mathlib's `taylor_isLittle0_univ` form then yields `vonMisesSRatio_second_order`: $E(a) - (\frac{1}{2} - a^2/16)$ is $o(a^2)$ at the origin. The remainder is $o(a^2)$ and not $O(a^4)$; the stronger form would need a fourth derivative and nothing here requires it. `coherent_solution_critical_exponent` then combines the expansion with `coherent_iff_sRatio_eq`: writing $a = Kr/D$ and rearranging $E(a) = D/K$ gives $K - 2D = Ka^2(\frac{1}{8} - 2q(a))$ with q the normalized remainder, and letting $K \rightarrow 2D$ along any family of positive coherent solutions with $r \rightarrow 0$ sends $q \rightarrow 0$ and $r^2/(K - 2D) \rightarrow 1/D$. The statement is deliberately filter-indexed rather than parameterized by a curve, so it constrains every such family and privileges no path. Because its hypotheses hold only eventually along the filter,

and an eventual clause is vacuously true on the empty one, the file also exhibits a family satisfying it: `coherentBranch` selects the unique positive solution above threshold via `supercritical_fixed_point_existsUnique`, `coherentBranch_tendsto_zero` reads `coherent_branch_continuous_at_threshold` as a limit along $K \downarrow 2D$, and `coherentBranch_critical_exponent` discharges every hypothesis on it. Scope is inherited unchanged from the file above: this is asymptotics of the solutions of a stationary equation. It establishes no stability, no dynamical selection and no time course, so reading it as a square-root foot on a recovery curve requires the adiabatic separation and a threshold crossing at non-zero speed, both of which the main text states as assumptions. Every declaration in the file depends only on `propext`, `Classical.choice` and `Quot.sound`.

The gap between these stationary-density results and any statement about a finite trajectory is the dynamical mean-field limit — propagation of chaos for the Kuramoto system, which is a research programme rather than a lemma [?, ?]. No theorem in this development closes it.

4.1 Finite mean-field trajectory controls

Two finite- N controls probe this gap without closing it. First, 500 seeded replicas of 1,000 identical-frequency oscillators were integrated at fixed coupling. Their steady order was 0.06266 below threshold, 0.16049 at threshold and 0.67735 above threshold. The last value lies -0.00535 from the stationary coherent value 0.68270. Early-growth fits gave $\lambda = 0.43240K - 0.82318$, against the infinite- N linearization $\lambda = 0.5K - 1$. This exhibits finite sampled escape toward the coherent branch; it proves no stability or dynamical-selection theorem.

Second, 1,000 ensembles were swept from $N = 10$ to $N = 1000$ at $K = 1$ and $K = 3$. In the subcritical control, mean order fell from 0.3726 to 0.0402 and absolute circular pair correlation from 0.0458 to 0.0010. The matched-sample sliced joint-product distance changed only from 0.0468 to 0.0419, exposing its finite-sample floor. Above threshold, mean order stabilized at 0.7208, but the unconditional lab-frame pair correlation remained between 0.459 and 0.525 rather than following the $1/N$ reference. The unpinned collective orientation is a shared latent variable, so unconditional independence and independence conditional on that orientation are different claims. In the co-rotating frame, the $N = 1000$ density at measured concentration $a = 2.1624$ had L^1 error 0.1291 from the von Mises target and Bessel self-consistency residual 0.00191. These finite, single-seed diagnostics prove neither form of propagation of chaos and discharge no Lean obligation.

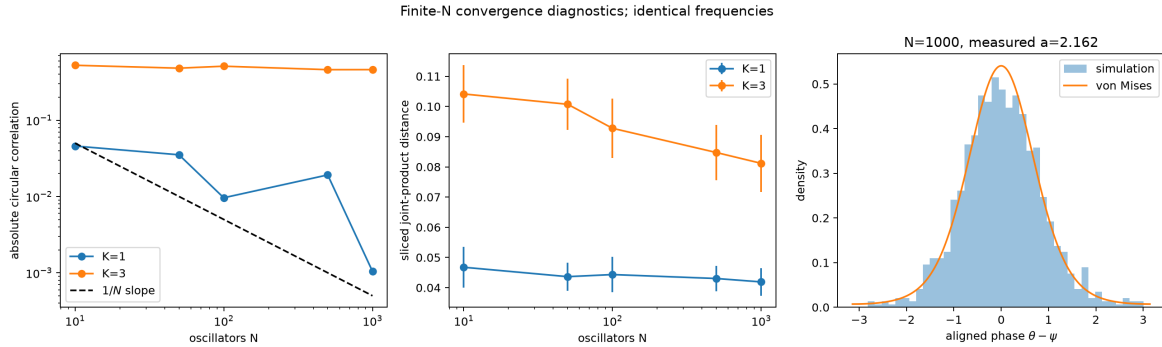


Figure 1: Finite- N convergence diagnostics for identical frequencies. The left panel compares branch-cut-invariant pair correlation with a $1/N$ reference; the centre uses matched sliced-Wasserstein joint-product samples; the right compares the co-rotating $N = 1000$ histogram with the von Mises density at measured concentration. The persistent supercritical lab-frame dependence is a negative result for the unconditional protocol, not a theorem about the conditional mean-field limit.

Perceptual anomalies such as visual snow can be modeled as boundary conditions where microscopic thermal noise (D) fails to be fully coarse-grained away by the continuous field, leaking across the K_c threshold into the macroscopic gestalt.

Lean 4 Implementation (Phase4RotatingFrame.lean): The potential above omits the natural-frequency term $-\sum_i \omega_i \theta_i$ that appears in the Lyapunov descent result `dV_dt_le_zero` of `Phase3.CombinatorialThermody`. The omission is not cosmetic. With any $\omega_i \neq 0$ the full potential is unbounded below (`kuramoto_potential_unbounded_below`), so it has no minimum for a phase-locked state to attain, and the two results concerned different functionals. This file chains them by the standard rotating-frame reduction $\theta_i \mapsto \theta_i - \Omega t$. For a system of identical

natural frequencies $\omega \equiv \Omega$: `is_kuramoto_trajectory_rotate` sends a trajectory to a trajectory of the zero-frequency system `sys.reduced` — the same system Theorem 5’s `ThermodynamicCover` works with; `kuramoto_potential_reduced` shows the two potentials coincide on it; `dynamic_potential_descent` transports the descent identity $\dot{V} = -\sum_i \dot{\theta}_i^2$ onto the potential above; and `rotating_frame_chain` runs this through the minimum characterization to phase-locking and $r^2 = 1$. Phase differences, phase-locking and the order-parameter magnitude are proved frame-invariant (`rotate_sub`, `is_phase_locked_rotate`, `order_parameter_r_sq_shift`), so no observable is lost. `Examples.lean` §7 runs the chain end to end on a two-oscillator system, discharging every hypothesis, and exhibits a system whose full potential has no minimum. Scope: the reduction is exact only for identical frequencies. A spread of frequencies leaves residual detunings $\omega_i - \Omega$ in the reduced system, to which the unboundedness result still applies, and phase-locking then depends on K_c , which this file does not treat — see the `Phase8_SelfConsistency.lean` note above for what is proved about the threshold, and note that it concerns a stationary density rather than the trajectories this file transports.

Trajectories that run. Descent along a trajectory is worth little if the only trajectory in the development starts at its own limit, which was the case until §7’s rigid rotation was joined by the results below. `kuramotoField_lipschitz` (`Phase3_CombinatorialThermodynamics.lean`) proves the Kuramoto vector field globally Lipschitz with constant $2\sum_{ij} |A_{ij}|$ — `sin` is 1-Lipschitz and the index set is finite, so there is no smallness caveat and the natural frequencies drop out. `is_kuramoto_trajectory_unique` turns this into Picard–Lindelöf uniqueness on all of $\mathbb{R}$: two solutions agreeing at one instant agree forever, in both time directions. Existence is now proved in general, and the gluing that Mathlib does not supply is written out. Mathlib’s `IsPicardLindelof` caps a local solution’s lifetime by the radius of the ball on which the field’s bounds hold, divided by the field’s sup norm there, and carries no global-in-time theorem; but `kuramotoField_norm_le` bounds this field by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$ on the *whole* state space, so the cap can be defeated by enlarging the ball after choosing the window, and `kuramoto_exists_on_window` produces a solution on every $(t_0 - T, t_0 + T)$. Uniqueness on an interval makes the family coherent — solutions on nested windows agree where both are defined — so $\theta(t) := \alpha_{[t-t_0]}(t)$ is a solution on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`), and `kuramoto_cauchy_problem` states existence and uniqueness together as a single `ExistsUnique`: the trajectory predicate is inhabited exactly once for each initial state, rather than only by the solutions we write down by hand. What existence does not supply is behaviour, and for that a concrete solution is still what we have. On two oscillators with unit coupling and zero natural frequency the phase difference obeys $\dot{\Delta} = -2\sin \Delta$, whose solution $\Delta(t) = 2\arctan(ce^{-2t})$ splits symmetrically into an exact trajectory of the full system (`Examples.lean` §15, `pairRelax_is_trajectory`). At $c = 1$ it starts a quarter turn out of phase (`pairRelax_not_locked_at_zero`); the cosine of the phase difference tends to 1, the order parameter tends to 1 through the two-oscillator identity $r^2 = (1 + \cos \Delta)/2$, and the dynamic potential tends to its global minimum. By the uniqueness result it is the only solution through its initial state. Two general statements about the long run are now proved as well, both for a zero-frequency system, which by `kuramoto_potential_unbounded_below` is the only case in which the potential is bounded below at all. `dynamic_potential_antitone` upgrades the instantaneous descent to monotonicity in t — `dynamic_potential_deriv_nonpos` constrains each instant separately and does not by itself say the trajectory settles — and with the uniform lower bound $-\frac{1}{2}\sum_{ij} |A_{ij}|$ monotone convergence gives `dynamic_potential_tendsto`: the potential converges along *every* trajectory. Integrating the Lyapunov identity gives the quantitative form, `dissipation_integral_eq`: energy dissipated over $[0, T]$ is exactly the potential dropped, so by `dissipation_integral_tendsto` the improper integral $\int_0^\infty \sum_i \dot{\theta}_i^2 dt$ converges, to the total drop. This needed the Lyapunov identity in `HasDerivAt` form rather than as a statement about `deriv`, which carries no differentiability information and cannot be fed to the fundamental theorem of calculus. `rotating_frame_dissipation` transports both to a uniform-frequency system. Finiteness of that integral is not the same as $\sum_i \dot{\theta}_i^2 \rightarrow 0$: an integrable function can spike forever on ever narrower intervals. The classical bridge is Barbalat’s lemma, which Mathlib does not have, so it is proved here as `tendsto_zero_of_lipschitz_of_integral_le`: a non-negative Lipschitz function with $\int_0^T g$ bounded above tends to zero, because a spike of height ε drags a window of width $\varepsilon/2K$ up with it and only finitely many such windows fit under a finite integral. Its hypothesis is then checked for the dissipation rate without differentiating it: the velocities are bounded by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$, the field is Lipschitz in the configuration, and the configuration is Lipschitz in time because its derivative *is* the bounded field (`trajectory_dist.le`, by the mean value theorem, with no invariant region needed since the bound holds on the whole state space). The conclusion is `velocity_sq_tendsto_zero`: along *every* trajectory of a zero-frequency system, $\sum_i \dot{\theta}_i^2 \rightarrow 0$; equivalently each $\dot{\theta}_i \rightarrow 0$ and the vector field along the trajectory tends to 0 in the state space. This is the first asymptotic statement here about an arbitrary trajectory rather than about a witness, and

Examples.lean §15 fires it on the relaxing pair, whose velocity starts at -1 and is therefore genuinely decaying rather than identically zero. It does not locate the limit. Locating it looks as though it needs the LaSalle invariance principle, which Mathlib does not carry, and a compact invariant set, which the state space $V \rightarrow \mathbb{R}$ does not supply; neither is in fact required. LaSalle is what one needs when a decreasing functional is all one has, and here the flow is a gradient flow of the dynamic potential — ∇_i of `kuramoto_potential_dynamic` is exactly $-\text{kuramoto_velocity}$ at zero frequencies, which is why `dynamic_potential_hasDerivAt` gives $\dot{V} = -\sum_i \dot{\theta}_i^2$ as an identity — whose minimisers are already characterised. `lojasiewicz_estimate` adds the missing ingredient: on the region where every pairwise phase difference lies within a quarter turn, $2a^2W \leq \left(\sum_{ij} A_{ij}\right) \sum_i \dot{\theta}_i^2$, where W is the excess of the potential over its minimum and $a > 0$ is a lower bound for the coupling. Its proof uses no spectral gap and no convexity, only two extremal indices: at the site m carrying the largest phase every coupling term has $\theta_j - \theta_m \in [-D, 0]$ with $D \leq \pi/2$ the spread, so every term is non-positive, and the term from the site carrying the smallest phase is at most $-a \sin D$ (`velocity_max_le`); against $W \leq \frac{1}{2} \left(\sum_{ij} A_{ij}\right) (1 - \cos D)$ the estimate follows from $\sin^2 D = (1 - \cos D)(1 + \cos D)$ and $\cos D \geq 0$, which is the only place the quarter-turn hypothesis is used. The compactness objection is met without any invariant region put in by hand. Because all terms of the excess are non-negative, a single pair satisfies $\frac{1}{2}a(1 - \cos(\theta_j - \theta_i)) \leq W$, so an initial excess below $a/2$ — which `dynamic_potential_antitone` preserves — keeps every pairwise cosine strictly positive for all time. A difference escaping $[-\pi/2, \pi/2]$ would have to cross $\pi/2$ in absolute value by the intermediate value theorem, and there the cosine vanishes; so the quarter-turn region is forward invariant (`spread_le_of_small`). This is branch-pinning by continuity on a connected half-line, and it replaces the quotient by the phase-shift symmetry we had thought necessary. The differential inequality $\dot{W} \leq -cW$ with $c = 2a^2/\sum_{ij} A_{ij}$ then integrates, in the elementary form “ $W(t)e^{ct}$ has non-positive derivative”, to $W(t) \leq W(0)e^{-ct}$ (`excess_decay`). The rate transfers to the velocity through $|\sin d| \leq \sqrt{2(1 - \cos d)}$, giving $|\dot{\theta}_i(t)| \leq Be^{-ct/2}$ (`velocity_abs_le_exp`), which is integrable on $(0, \infty)$; since $\theta_i(t) = \theta_i(0) + \int_0^t \dot{\theta}_i$, the trajectory converges (`phase_tendsto`), and $W \rightarrow 0$ forces the limit to be phase-locked, hence a global minimiser with $r^2 = 1$ (`kuramoto_tendsto_global_minimum`). `rotating_frame_tendsto_global_minimum` transports it to uniform frequencies, where what converges is the relative configuration. The two hypotheses are on the initial data and neither can be removed: splay and twisted configurations are equilibria too, so “every trajectory reaches the phase-locked state” is false, and what is proved is the arc condition version — which the two-oscillator witness satisfies for every c , since $2 \arctan$ lands in $(-\pi, \pi)$. **Examples.lean** §17 runs the theorem on three sites, where the method of §15 is unavailable because the system has no closed-form solution: the trajectory is produced by `is_kuramoto_trajectory_exists` from the initial data $(0, 0, \frac{1}{2})$, which is not phase-locked, and its limit by the convergence theorem as $\theta(0) + \int_0^\infty \dot{\theta}$; the numerical content of checking the hypotheses is $\cos \frac{1}{2} > \frac{3}{4}$, supplied by `Real.cos_bound`.

From the continuous field to the discrete model (`Phase2SimplicialBridge.lean`, `Phase2MeshConvergence.lean`):

The coupling matrix A_{ij} above is not postulated. A triangulation of the substrate assigns to each edge (u, v) a region of the manifold, and the coupling weight is the Lebesgue–Bochner integral of the scalar magnitude of the stress-energy tensor over that region (`edge_weight`); symmetry and non-negativity of A are then proved rather than assumed. The bridge back is mesh refinement: as the regions shrink, the discrete energy $\frac{1}{2} \sum_{u,v} \mu(R_{uv}) f(x_u)$ should converge to the continuous energy $\int_S f$. Stated as a conjecture over the bare triangulation class, that is false: nothing there forces the regions to cover the manifold, empty regions have diameter zero, and the everywhere-empty triangulation refutes it. Stated correctly it is a theorem (`mesh_refinement_convergence`), quantified over a sequence of triangulations of a region S whose regions are measurable, pairwise disjoint across distinct edges, covering, and anchored to the embedded vertices, with mesh size tending to zero. The fixed-mesh error is bounded by $\varepsilon \mu(S)$, with ε a modulus of continuity of the integrand, and the symmetric double sum is proved equal to the midpoint rule over unordered edges. Two limitations are worth stating. The theorem gives convergence, not the $O(1/N^2)$ rate reported by `simulations/mesh_refinement.py`. And `TriangulatedManifold` itself is bare data: it pairs a complex, an embedding and a family of edge regions without relating them, so the geometric conditions live in a separate predicate `IsRegularTriangulation`. Because those conditions are in tension — cells disjoint yet covering, every vertex anchored to each of its edges — **Examples.lean** §6 exhibits an instance: the uniform partition of $[0, 1)$ into N half-open cells, on which the convergence theorem is instantiated and the approximations are checked to genuinely change with N .

The discretization structure `DiscreteThermodynamics` now *carries* that predicate as a field, which is

what makes its edge weights integrals over a geometrically constrained family rather than over arbitrary sets, and three statements follow that were unavailable before. `weight_self`: a vertex has no edge to itself, so the coupling matrix has zero diagonal. `face_of_weight_ne_zero`: a nonzero weight $w(u, v)$ forces $\{u, v\}$ to be an actual face of the complex, so the discretization cannot invent couplings between vertices the triangulation does not join. And `total_weight_eq_setIntegral`: $\frac{1}{2} \sum_u \sum_v w(u, v) = \int_S |T|$, i.e. the total coupling strength is exactly the scalar stress-energy of the region covered — nothing counted twice (disjointness), nothing dropped (covering). That last identity is the precise sense in which the coupling matrix is derived from the continuum rather than posited, and it is exactly what the everywhere-empty triangulation violates. `Examples.lean` §6 witnesses the strengthened structure on the same grid: each consecutive edge carries weight $1/N$, and the weights sum to 1 for every N . The witness is one-dimensional and its stress-energy tensor is constant, so it establishes satisfiability, not a physically interesting discretization. One field of the structure is deliberately left free and is now marked `[MODELLING]: scalar_magnitude`, the scalar read off the stress-energy tensor, constrained only to be non-negative. A $(0, 2)$ tensor has no canonical scalar — the trace, an observer’s energy density and the tensor norm are different functions, and the structure carries neither a metric nor a frame — so imposing one would pin a choice rather than derive it, and the three statements above hold for every non-negative choice, so it would prove nothing new. That the choice is nonetheless a physical input is stated as a theorem rather than left as a remark: §6 exhibits a second instance over the same triangulation, tensor and regions, differing only in that field, for which the partition identity again holds while every edge weight differs (`gridThermo_magnitude_matters`). The remaining constraint, non-negativity, is the discrete shadow of an energy condition and is what makes the induced couplings a legal Kuramoto coupling.

4.2 The fold-and-cross proof that E is strictly decreasing

Uniqueness of the coherent branch is the injectivity of $E(a) = \mathbb{E}_a[\sin^2 \theta]$ on $(0, \infty)$, and what is proved is the stronger statement that E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_strictAntiOn`). The main text states the conclusion and the shape of the argument; this is the argument, and it avoids the route we had expected to need. Differentiating the exponential family in its natural parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation under the integral sign and leaves a covariance that is not sign-definite pointwise: symmetrising with an independent copy gives $\frac{1}{2} \mathbb{E}[(X - X')^2(X + X')]$ with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the proof differentiates anything. The target is kept as the two-point inequality $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$ that the covariance form needed the tilt to perform, done once at the start; its analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$ with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; evaluating at $y = x_0$ and integrating in x alone gives the result, since the term carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by — and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$, the monotonicity of the Bessel ratio, which Mathlib carries in no form.

5 Global Sections and the Unity of Consciousness

Phase synchronization alone is mathematically insufficient for conscious unity; the synchronized local fields must be topologically glued together.

Theorem 5 (Sheaf-Theoretic Unity): Given a sheaf of probability measures $\mathcal{F}$ over a continuous biological substrate X , if local states $P_U \in \mathcal{F}(U)$ agree on all spatial intersections $U \cap V$, they uniquely glue into a Global Section $P_X \in \mathcal{F}(X)$. The overlap agreement is an independent physical hypothesis; phase-locking does not imply it.

Lean 4 Implementation (Phase5_GlobalSection.lean): The headline gluing theorem separates two inputs. The class field `ThermodynamicCover.thermodynamic_equilibrium` places the phase field at the potential minimum, while `LocalSectionSynchronization.section_agrees_of_phase_eq` independently requires equal-phase patches to agree on overlaps. Only the second supplies the compatible family consumed by the sheaf theorem; no dynamics in the development derives it. What the second one says can

be stated without the sheaf: on a discrete substrate two patch sections have the same restriction to their overlap exactly when they assign the same mass to every site they share (`restrict_eq_iff_densityOn_eqOn`), so the hypothesis is that patches at a common phase carry a common local density where they meet. It excludes data the rest of the structure admits: a two-patch cover of the three-site substrate at one phase, with profiles differing at the shared site, satisfies every other requirement and fails this one (`overlap_agreement_fails`). It does not require patches to agree away from their overlaps — the three-patch witness carries three pairwise-different profiles. The equilibrium field is discharged one file over on a class of initial data: `Phase5.EquilibriumBridge.lean` supplies `ThermodynamicCover.ofConvergentTrajectory` from `kuramoto_tendsto_global_minimum`, and `Examples.lean` §17.1 is a three-patch cover built over the resulting phase field. The cover class also has an independent instance (`Examples.lean`, §4), so the theorem is not conditional on a structure of unknown realizability. Its three-site substrate has two overlapping patches meeting at one site; local sections are germ families of the phase-indexed measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ in the sheafification of the probability presheaf. The family is provably non-constant in θ (`phaseMeasure_not_const`), and `thermodynamic_equilibrium` follows from `phase_locked_minimizes_potential`. This is a formalized gluing mechanism under an explicit compatibility postulate, not a derivation of compatibility from synchronization. Three results sharpen its scope. First, `potential_min_iff_phase_locked` (`Phase4.KuramotoDynamics.lean`) characterizes the minimizers of the reduced potential in both directions, from which `ThermodynamicCover.phase_locked` follows for every instance. Second, `Examples.lean` §13 carries a non-constant phase field and a non-uniform invariant measure, with patch-local densities read from the sections through `densityOn`; agreement at the shared site is an equation between separately computed numbers. The limit is structural: phase-locking equates phases, while agreement of the associated sections remains the independent class field named above.

Third, and this one changed the theorem rather than the witnesses. As stated until 2026-08-30, Theorem 5 in Lean was uniqueness of the global state given the local data, not emergence of a global state from it: the class field `phase_invariant_measure` handed each instance a section over $\mathbb{T}$, and `sync_to_section_eq` declared each local section to be a restriction of it, so the existence half was assumed and the sheaf condition only excluded competitors. Both fields have been removed. `LocalSectionSynchronization` now constrains its local data only by `section_agrees_of_phase_eq` — patches at a common phase agree where they overlap — in which every symbol ranges over one patch or one overlap. The sheaf property is isolated as `probability_glue_unique`, a lemma with no physics in it, which takes a compatible family to the unique section over $\mathbb{T}$ restricting to it; `ThermodynamicCover.invariantMeasure` is defined as that section, and the statement that every local section is its restriction is a theorem about it, `ThermodynamicCover.sync_to_section_eq`. An instance may still be built the other way round, from a global measure, through the constructor `LocalSectionSynchronization.ofInvariantMeasure`, and §4 and §13 are built through it — proof that the weaker class is genuinely weaker, and the reason those two witnesses glue to the measure they started from. `Examples.lean` §14 is built without it: the two patches carry independently chosen mass profiles (2, 1, 7) and (5, 1, 3), agreeing only at the shared site where both put mass 1, and the section the gluing returns has profile (2, 1, 3). It is neither profile, and neither profile restricts correctly to both patches (`leftW_glues_nothing`, `rightW_glues_nothing`), so the global state was produced by the theorem and not supplied to it. One caveat on that witness has since been removed: its patch-local sections were originally *defined* by restricting a measure on the whole substrate, because the germ-measure dictionary of §10 was built at $\mathbb{T}$ only. That dictionary is now stated at an arbitrary open — `densityOn` reads a section over any U , `sectionOfMassOn` builds one there, and `massEquivOn` is the resulting bijection with the mass profiles on U — so each patch’s section is constructed over its own patch and no measure on the whole substrate appears in the instance. `glued_section_eq_restrict` records that the object is unchanged, so the construction and not the content is what moved. What remains assumed is the overlap agreement itself, and `thermodynamic_equilibrium`: no dynamics is run into the minimum.

6 Reflexive Topology and the Self

To minimize internal thermodynamic friction, the unified Global Section must model its own internal state changes.

Theorem 6 (Conditional Auto-Resonance): On a complete metric space of global sections, a self-prediction map has a unique fixed point if it is Lipschitz at the independently assumed rate $e^{-(K-K_c)\tau/2} < 1$. Coherence supplies $K > K_c$; it does not supply the Lipschitz law.

Lean 4 Implementation (Phase6.ReflexiveTopology.lean): We formalize the reflexive projection as a contraction on a complete metric space and apply the Banach fixed-point theorem to obtain a unique

fixed point — the “Self” as a topological feature rather than a metaphysical addition. The entropy production functional does not appear in this file: the identification of the Banach fixed point with the minimizer of σ is an interpretive step made in the main text, and the Lipschitz bound at the mean-field rate is an instance obligation rather than a consequence of field dynamics. Without an instance of `ReflexiveBoundary` and a metric on `GlobalSection`, the theorem would be conditional on structures of unknown realizability — the same failure mode an uninhabited class has. `Examples.lean` §10 supplies both on the three-site cortex of §4, and the metric is built from the measures rather than imposed. Five stages. (i) The germ-measure dictionary: on a discrete substrate every point has a smallest open neighbourhood, so “mass carried at x ” is a well-defined map out of the stalk at x (constructed as the colimit map of a cocone whose cocone law is precisely that restriction preserves point masses); reading a section’s germ at each site of an open U gives a map to mass profiles on U , and `massEquivOn` proves it a bijection — injective because a germ on a discrete space is determined by its restriction to the point, surjective because every prescribed profile is realized by a finite measure on U . `massEquiv` is the case $U = \mathbb{T}$; the general case was added on 2026-08-30 and is what lets Derivation 5’s §14 witness build its patch-local sections over their own patches. Mathlib supplies nothing here: `TopCat.Presheaf.sheafify` has no adjunction and no `isIso.toSheafify`, so the inverse had to be built. (ii) The metric: the distance between two sections is the largest difference between the masses their measures put on a site, i.e. the uniform distance between their densities (equivalent to the total variation of the glued measures on a finite substrate, within a factor of the number of sites, but not equal to it). It is complete, proved by exhibiting the limit section of a Cauchy sequence of densities, and `exists_dist_lt_one` exhibits distinct sections at distance $1/2$. (iii) The Self, built through the avatar (rebuilt 2026-08-31). Until that pass `ReflexiveBoundary` carried the predictive map as free data beside the avatar, and `reflexive.topology.implies.self` projected that map out without ever touching the avatar; the consequence was a theorem, `self_of_constResonance`, stating that replacing the avatar by a constant left a legal boundary with the same predictive model and the same Self. The structure now carries the avatar’s *write* (`auto_resonance`) and its *read-out* (`readout`), and `predict` is their composite rather than a field, so the self-model runs through the avatar region or it does not exist. On the cortex the map is `avatarRead` — the single mass a section over the one-site avatar region carries — followed by `avatarReadout`, which returns that mass averaged with the baseline; `cortexPredict.dist` gives the exact contraction factor (one half of *what the avatar sees*: the read-out is blind off the avatar region, and that loss is the fold, stated rather than hidden), `cortexPredict.not_const` that it is not constant, `cortexPredict.fixed.unique` that its fixed point is unique. The previous witness `relax`, which averaged the whole field with the baseline, could not be kept: it reads every site and so does not factor through a one-site avatar. The earlier 0/1-metric witness is kept as `gsDiscreteMetric`, together with `contracting.implies.const`, which proves that under that metric every contraction is constant — the record of why the earliest witness said nothing. (iv) The avatar as a constraint. Following the rule that data classes stay bare, the resonance condition is a predicate and not a field: `IsRestrictionResonance rb` says `rb.auto_resonance s` is the presheaf restriction of s to the avatar region. Four theorems fix its status. `eq.of.avatar.eq`: if the avatar regions of a family of resonant boundaries cover the substrate, two global states with the same reading at every avatar are equal — the readings are local, and the global state is reconstructed from them by `probability.glue.unique`, the same sheaf condition Derivation 5 uses. `self.mem.range.readout`: a fixed point lies in the range of the read-out, so the Self is reconstructed from a single section over the avatar region — the low-dimensional fold, as a theorem. `self.eq.readout.restrict`: under resonance the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$, the field being what its own localized self-encoding reconstructs; this is the statement a structure carrying `predict` as free data cannot make, since there `predict` is unrelated to the restriction. And `not.self_of_constResonance` is the negative half: the blinded boundary’s map is constant, `constResonance.existsUnique.self` names its only fixed point as the state its stipulated contents extend to, and the original Self is not one. The opposite statement, that blinding leaves the Self where it was, does not typecheck against this structure — a contraction hypothesis about a boundary’s own map says nothing about the blinded one’s — and `constResonance.not.isRestrictionResonance` shows the blinded boundary fails the predicate as soon as two global states differ on the avatar region. On the cortex all of it is exercised: `cortexReflexive` is built from the restriction and discharges the predicate by `rfl`; `cortexReflexive.restrict.ne` checks the avatar region separates the silent field from the baseline, so the predicate is not vacuous there; `siteReflexive` gives one avatar per site, and since the singletons cover, `cortex.eq.of.avatar.eq` reconstructs the global state from three local readings; `cortexBlind.self.ne` proves the baseline is *not* a fixed point of the blinded map, and `cortexBlind.not.determined` exhibits two distinct states with identical blinded readings. Banach’s uniqueness half, which the original statement did not carry, is

`self.unique`. (v) The contraction rate, from K and D . `resonanceRate` $K D \tau = \exp(-(K - K_c)\tau/2)$ with $K_c = \text{critical_coupling}$ $D = 2D$ is the linear relaxation factor of the mean-field order parameter over an interval τ . `resonanceRate_lt_one` and `one_le_resonanceRate` split the parameter plane at Sakaguchi’s threshold; `self_of_supercritical` derives the unique fixed point from $K > K_c$, and `Chain.lean`’s `self_of_coherent_order_parameter` derives it instead from the *existence* of the coherent order parameter — a distinction that matters, since K_c is a definition unfolding to $2D$, so the first statement is insensitive to whether the bifurcation is proved and the second is not, and `not_contractingWith_resonanceRate` records that at or below threshold `ContractingWith` at that rate is false for every self-map of every metric space, so the argument is unavailable rather than merely unproved. The witness sits at $K = 3$, $D = 1$, $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out achieves (`cortexResonanceRate`), and `cortexSubcritical_not_contracting` runs the negative half on the same map at $K = 1$. Three limitations remain and we state them. The read-out `avatarReadout` is a modelling choice: no field dynamics in the development produces it, and the Lipschitz bound at the mean-field rate is an instance obligation, not a theorem — what changed is that the obligation is now stated in the substrate’s parameters and its consequence is conditional on the threshold. The identification of this three-site substrate’s coupling with $K = 3$ is a choice of units for the witness and not a measurement. Nothing derives the resonance condition either: that a physical avatar performs the restriction is an assumption on an instance, in the same class as the overlap-agreement obligation of Derivation 5. And the Lévy–Prokhorov metric named in the file header is *not* what is constructed: on a discrete substrate the ε -thickening of a set is the set itself for $\varepsilon < 1$, so $\mu \mapsto \frac{1}{2}\mu + \frac{1}{2}\mu_0$ is not a Lévy–Prokhorov contraction there, and Mathlib carries no completeness result for that metric. The uniform distance between densities is the metric this substrate supports; we make no claim that it and Lévy–Prokhorov are equivalent.

7 Topological Rigidity of von Neumann Architectures

Finally, we compare the continuous coupling matrix $A(x, y)$ of biological fields with the discrete, rigid routing matrix G_{ij} of standard artificial hardware (e.g., GPUs).

Theorem 7 (Hardware Thermodynamics): Because a continuous parameter space contains global minima inaccessible to a discrete lattice under identical thermodynamic constraints, the dynamical potential of a biological neural field is strictly lower than that of rigid silicon hardware: $V(\text{continuous}) < V(\text{discrete})$.

Lean 4 Implementation (Phase7HardwareComparison.lean): The lemma `exists_better_coupling_allocation`, wrapped as the theorem `continuous_beats_rigid_topology`, proves the reallocation step: given a coupling matrix satisfying `is_strictly_suboptimal` (positive weight on a less phase-correlated pair while a more correlated pair exists), there is another valid coupling with identical total resources and strictly greater total phase correlation.

Lean 4 Implementation (Phase7Rigidity.lean): Two defects of the above are addressed here. (i) *Rigidity as a support constraint.* `RealizableIn S R A` says A is a valid coupling meeting resource budget R and vanishing on every pair the architecture has no wire for — weights tunable, wiring fixed. `totalCorrelation_le_of_realizable` proves no realizable coupling exceeds $R \cdot M$ when M bounds phase correlation over the wired pairs, and `exists_realizable_pair` attains it. `rigid_is_strictly_suboptimal` then *derives* `is_strictly_suboptimal` for every realizable coupling, whenever some pair (k, l) the wiring misses has correlation exceeding M : the hypothesis Theorem 7 would otherwise have to assume a consequence of the wiring pattern. `rigid_gap` gives the quantitative form, a gain of at least $R(\cos(\theta_l - \theta_k) - M) > 0$ at equal budget. `Examples.lean` §8 witnesses this on three sites wired only to the maximally anti-correlated pair, where the gap is exactly 2, and runs `exists_better_coupling_allocation` on it with no assumed hypothesis. (ii) *Continuity as a measure-theoretic distinction.* `fieldCorrelation` integrates a kernel against $\mu \otimes \mu$ as in Derivation 4. A finitely-sited architecture is a `Finset X`; `fieldCorrelation_sited_eq_zero` proves that on an atomless substrate such a kernel contributes exactly zero, for every phase field and every choice of weights, while `fieldCorrelation_patchKernel` gives $c\mu(U)^2$ for a patch. `sited_architecture_below_field_optimum` combines them; `Examples.lean` §8 instantiates it on $\mathbb{R}$ with a thousand units of weight at the origin against the unit interval.

Scope caveat. Result (i) is resource-matched and not question-begging, but it turns on fixed wiring support, not on continuity: an architecture wired to the best pair is not beaten. Result (ii) is a genuine type-level separation but is *not* resource-matched, since a measure-zero substrate carries zero resource in the $\mu \otimes \mu$ sense as well; what it establishes is that the continuum coupling energy cannot register a finitely-sited architecture at all, which is a claim about the model’s observable rather than about

computational power. Neither result formalizes manifold structure or lattice geometry. Accordingly, the inference to “von Neumann architectures cannot experience unified consciousness” remains an informal argument in the main text, now with two verified steps rather than one.

8 Claims and their Lean identifiers

The table below is the claim-by-claim map, kept here rather than in the main text. It is here because it is a reference rather than an argument: each row names the Lean identifier that carries a claim and states the scope limitation of that identifier, and no step of the main text depends on the reader working through it. The six-row summary in the main text (Table 1) is the argument; this is the audit trail behind it.

Table S1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. A physical postulate is an obligation on each instance rather than a global assertion, which is the shape a standalone axiom cannot have (Section 9); two of the postulates a reader might expect to find are not postulates at all, but consequences of the local data (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Capacity of a finite phase space	The first premise, discharged by theorem rather than by footnote (<code>Phase1.PhaseSpaceCapacity</code>). $H(p) \leq \log X $ for any distribution on a finite state space, with equality exactly at the uniform law (<code>shannon_entropy.eq_log_card_iff</code>); a stream with $ P ^k > X $ histories cannot be recorded injectively (<code>absorb_not_injective</code>) and its entropy exceeds anything the system can hold; and on a <i>finite</i> space an unreachable state is an erasure, hence heat (<code>finite_phase_space.dissipates</code>). Finiteness is assumed, not derived, and is fenced — <code>Nat.succ</code> refutes the last step on an infinite space (<code>Examples §19</code>). The symmetry group and the metric are still consumed by no theorem	Theorem
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (<code>Phase1</code>), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (<code>Phase3</code>); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (<code>Phase1</code>); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Inevitability of Boundaries (π_0)	The domain-wall case of the Kibble mechanism (<code>Phase1 §3</code>): on a connected substrate, a continuous field taking values in two different connected components of the vacuum manifold must leave the vacuum somewhere (<code>exists_notMem_of_connectedComponentIn</code>). Witnessed on the double well, whose vacuum manifold is computed to be exactly $\{-1, 1\}$ and proved disconnected (<code>Examples §11.1</code>); the field class is non-empty and the forced wall is located. Higher homotopy (π_1 vortices, π_2 monopoles) available and not taken	Theorem
Nonpredictive information bounds dissipation	Still, Sivak, Bell and Crooks’ bound is the field <code>PredictiveDissipation.still_bound</code> (<code>Phase3.PredictiveThermodynamics</code>). Everything downstream is derived: mutual information is defined from Mathlib’s <code>kLDiv</code> ; the data processing inequality for the chain $X_t \rightarrow S_t \rightarrow S_{t+1}$ proves the nonpredictive information non-negative (<code>predictiveInfo.le_mutualInfo</code>), from which $W_{\text{diss}} \geq 0$ follows rather than being assumed; a dissipation budget bounds wasted memory, and zero dissipation forces every retained bit to be predictive. Witnessed in <code>Examples §18</code> by two systems over one two-bit law, one permitted to dissipate nothing and one forced to dissipate its whole memory, with a counterexample showing the no-back-action hypothesis is load-bearing. This replaces the earlier “gradient descent on σ drives $D_{KL} \rightarrow 0$ ”, which was an invalid inference. On a finite register with a bipartite environment the postulate is discharged rather than assumed: <code>PredictiveDissipation.ofLandauer</code> (<code>Phase3.LandauerBridge</code>) builds a predictive structure whose thermal scale and dissipated work are the register’s own and derives <code>still_bound</code> from <code>landauer_bound</code> , given the identification of wasted memory with erased entropy. That identification is the physical content and is not free — a reversible update erases nothing, so it forces the waste to zero (<code>nonpredictive.eq_zero_of_injective</code>) — and on the one-bit eraser the bound is attained (<code>Examples §18.5</code>)	Theorem + instance postulate
Structural Resonance KL Bound	Gibbs’ inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (<code>Phase3.KLBound</code>). Retained: the postulate is consistent and yields $\sigma \geq 0$. No limit may be read off it	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy-production-rate</code> . Not restricted to finite substrates: the drift map is a bounded operator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$ (<code>kernelCLM</code>), its adjoint turns the derivative into a gradient vector (<code>hasFderivAt.quadratic-grad</code>), and descent follows on any finite measure space (<code>structural.resonance.decreases.sigmaContinuum</code>), witnessed on a non-atomic substrate (Examples §20). The phase field is still held fixed: this is plasticity of the coupling, not joint (θ, K) dynamics	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase.locked.minimizes.entropy.of.symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is antisymmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total.drift.eq.of.symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region's stress-energy; mesh refinement converges, witnessed on $[0, 1]$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate
Coarse-graining of a symmetric coupling	<code>mesh.refinement.convergence</code> (Phase2) proves convergence of the scalar coupling energy along regular refining triangulations. On the formalized $[0, 1]$ grid the energy is the composite trapezoidal rule, so a C^2 integrand has error at most $\zeta/(12N^2)$. The result produces neither a kernel nor dynamics: <code>edge.region</code> supplies no product-structured data on $M \times M$. A genuine kernel limit requires new architecture before the separate dynamical mean-field problem can be stated	Theorem (scalar energy only)
The field is the endogenous EM field	Conditions formalized; the identification itself not derived. A continuum coupling kernel is posited after the scalar convergence result; identifying that kernel with cortical ephaptic coupling is the framework's central empirical commitment (Section ??). <code>IsEMFieldCoupling</code> (Phase9. <code>EMIdentification.lean</code>) is the predicate <code>E56</code> carries: a jointly continuous kernel, a normalised substrate, a coupling constant K equal to that kernel's double average, no single site contributing more than half of it, and a noise D that is the field's own. No object in this development denotes cortex, so whether cortex satisfies them is not a Lean question. Supported as <i>modulatory and integrative</i> , not as primary: measured endogenous fields of 1–5 mV/mm shift spike timing by 1–3 ms. Four falsification conditions are stated in the text	Modelling assumption + empirical commitment
Frustration as memory capacity	Outside the regime the chain covers. <code>ThermodynamicCover.A.pos</code> excludes frustrated couplings at the gluing node, and circle-valued offsets from patch phases are coboundaries. Cortical phase singularities supply integer transitions between local real lifts instead: <code>loopWinding.eq.zero.of.hasGlobalLift</code> proves non-zero winding obstructs a global lift, with a winding-one witness. No theorem connects winding sectors to content	Theorem (topological obstruction) + empirical candidate
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a <code>ThermodynamicCover</code> , whose fields are all local: the global section is constructed by <code>probability.glue.unique</code> , and the statement that every local section is its restriction, <code>sync.to.section.eq</code> , is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is not an assumption on every instance: <code>ThermodynamicCover.ofConvergent-Trajectory</code> (Phase5 <code>EquilibriumBridge</code>) discharges it from <code>kuramoto.tendsto.global.minimum</code> , and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i> . The overlap-agreement hypothesis is untouched by any dynamics and is read pointwise: two patch sections have the same restriction to their overlap exactly when they assign the same mass to every shared site (<code>restrict.eq.iff.density0n.eq0n</code>), and a two-patch cover at one phase with profiles differing at the shared site fails it while meeting every other requirement (<code>overlap.agreement.fails</code> , Examples §20)	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a self-prediction map (Phase6); metric, completeness and the Lipschitz law at $e^{-(K-K_c)\tau/2}$ are hypotheses, discharged on the three-site witness. Coherence yields $K > K_c$ and hence makes the stipulated rate less than one; it does not prove that the avatar read-out contracts at that rate. <code>E89</code> carries both the independent Lipschitz law and the identification of the glued state with that fixed point	Conditional theorem + modelling postulate (witnessed)

Conceptual Claim	Lean 4 Implementation	Status
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (<code>Phase4.RotatingFrame</code>); identical frequencies	Theorem
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (<code>kuramotoField.lipschitz</code> , <code>kuramotoField.norm.le</code>), and through every state at every instant passes exactly one trajectory, defined for all time (<code>kuramoto.cauchy.problem</code>). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (<code>dynamic.potential.tendsto</code>) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (<code>dissipation.integral.tendsto</code>). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (<code>velocity.sq.tendsto.zero</code>). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (<code>kuramoto.tendsto.global.minimum</code>), by a Polyak-Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (<code>Examples §17</code>)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (<code>Phase7.Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed.point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); that it does so at the mean-field rate $r^2/(K - K_c) \rightarrow 1/D$, i.e. $\beta = 1/2$ (<code>Phase8.CriticalExponent:coherent.solution.critical.exponent</code> , witnessed by <code>coherentBranch.critical.exponent</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)
Conditional composition	<code>Chain.lean</code> : <code>chain</code> takes eight named hypotheses and concludes <code>UnifiedSelf</code> , meaning that the particular section glued by a thermodynamic cover is the unique fixed point of self-prediction. E89 explicitly supplies both the Lipschitz law and the identification of the glued state with a fixed point; Banach supplies uniqueness, not that identification. <code>chain.hypotheses.jointly.satisfiable</code> discharges all eight on one toy substrate and proves only that the hypotheses are simultaneously consistent, not that one mechanism realizes them. E12 remains equivalent to its conclusion, so the first two links are independent roots. Two of the eight ask for a structure tied to the objects of their premise rather than for bare existence: E34 for a predictive structure carrying the register's own temperature and Landauer heat and a non-zero waste, E78 for a cover coupled at least at K whose equilibrium is the limit of a relaxation on that coupling (<code>ThermodynamicCover.IsReachedByRelaxation</code>); each is discharged in the joint witness and each has a regression rejecting a structure that would satisfy the bare form	Conditional theorem with eight named hypotheses
Coarse-graining produces no kernel	Negative result (<code>Chain.lean §9</code>). <code>ContinuousNeuralField</code> carries no conditions, so a constructor from a mesh limit would be free of content (<code>continuousNeuralField.free</code>). <code>mesh.refinement.convergence</code> converges a <i>scalar</i> , and <code>TriangulatedManifold.edge.region</code> is a subset of M rather than of $M \times M$, so the discrete side has no product structure to pass to a limit. The vertex-sited candidate does carry the weights (<code>vertexKernel.apply.embedding</code>) and still contributes exactly zero continuum coupling energy on an atomless substrate (<code>vertexKernel.fieldCorrelation.eq.zero</code>) — Derivation 8's hardware no-go, cutting against the framework's own coarse-graining step	Theorem (negative)

9 Formal soundness: why a global axiom is the wrong shape for a physical postulate

Formalization is only as trustworthy as the axioms it rests on, and an axiom that is inconsistent silently makes every theorem downstream of it vacuous while leaving the build green. We report a case of exactly this, found by auditing our own development, because the failure mode is easy to reproduce and we have

not seen it discussed in the applied-formalization literature.

The failure has one shape, and naming it is more useful than cataloguing its instances: *an axiom that constrains a symbol it does not itself bind*. Such a declaration is not a postulate about one system. It is a constraint on every model of the class the symbol belongs to, including models built to violate it, and a class that has such a model is thereby inconsistent. Three concrete cases, each of which proves **False** on its own:

1. **spontaneous_symmetry_breaking_pointwise_min** took the potential-energy functional as an *implicit* argument occurring only in its hypothesis **PotentialEnergy** $\phi = V \ v_0$. Instantiating that functional with the constant map $\lambda _. V(v_0)$ discharges the hypothesis by **rfl** and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$ this proves $1 = 0$.
2. **landauer_heat_eq** equated the free class field **heat_dissipation** with a bath-entropy difference *for every instance* of the class. An instance with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
3. **kl_bound_axiom** asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ is a single real fixed by a class field. Since D_{KL} is unbounded above, a point mass against a uniform distribution refutes it.

Two further declarations of the same shape are worth recording because their refutation is different in kind. **phase_invariant_periodic** and **sync_to_section_eq**, as global assertions pinning free fields of **LocalSectionSynchronization**, are refutable as soon as the probability presheaf admits two distinct global sections; the refutation needs no arithmetic at all, only that the sheaf is not trivial. Neither is a postulate in the current development: the periodicity condition is an obligation of the constructor that builds a cover from an invariant measure, and **sync_to_section_eq** is a theorem about the global section the sheaf condition produces.

The rule that prevents it is uniform: *a physical postulate that mentions a class field must be a field of that class, never a standalone axiom quantified over all instances*. A postulate is then an obligation each model discharges, not a global claim about every model. Under that rule the development declares no axioms at all.

9.1 What the witnesses establish, and what they do not

An uninhabitable class makes its theorems vacuous just as surely as an inconsistent axiom did. **Examples.lean** therefore constructs witnesses for every class that carries a physical postulate: a one-bit erasure system with a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging **kl_bound**; and a scalar field on a one-point spacetime discharging the action principles. The gluing structures are the most demanding case and are witnessed by a **ThermodynamicCover** over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two overlapping open patches whose intersection is a single shared site.

What the witnesses establish is that the classes are uninhabitable, so the theorems about them are not vacuous. What they do not establish is that a cortex is such an instance. In the cover above the phase field is already constant — every patch locked at $\theta = 0$ — which is precisely what makes the equilibrium obligation dischargeable; getting a physical system into that configuration is the content of the informal dynamical argument.

10 Empirical commitment: the endogenous electromagnetic field

The framework identifies its posited continuum coupling kernel with the endogenous electromagnetic field of mammalian cortex. This is a claim about biology, not a consequence of any theorem above it.

10.1 Supporting evidence

Neurons do not only signal through synapses: transmembrane currents generate extracellular fields that shift membrane potentials of neighbouring cells, a mechanism known as ephaptic coupling. Endogenous fields of 1–5 mV/mm entrain spike timing [?, ?], and stable electric fields carry information that survives representational drift in the underlying spiking population [?, ?, ?]. The field is spatially continuous and superposes contributions from large populations — a medium in which spatially separated activity

is combined without further mechanism. This is the integrative role the CEMI field theory assigns it [?, ?, ?], physically supported by astrocytic syncytia joined by gap junctions [?].

The endogenous field’s effect on any individual neuron is modulatory: it biases when a cell fires; it does not decide whether it fires [?]. The claim is therefore that the field is modulatory in effect and integrative in spatial reach — weak per synapse, and the only variable in the system that is defined everywhere at once. What the framework needs is the reach, not the strength.

10.2 Finite-sheet spatial-decay control

A finite stochastic control asks whether fixing total coupling while shortening its spatial reach fragments the phase field. On a periodic 128×128 sheet spanning 2.0 mm, an exponential kernel was row-normalised to total coupling $K_0 = 8$ at every decay length; the dynamics used thermodynamic diffusion $0.5 \text{ rad}^2/\text{s}$, quenched Gaussian frequency spread 1.5 rad/s , and 20,000 Euler–Maruyama steps per length. With steady $r > 0.2$ as the declared operational coherence criterion, the smallest interpolated decay length above which every larger sampled length remained coherent was 0.0140 mm. The transition-local trajectories were metastable and non-monotone, so that number is an operational boundary rather than a critical-point estimate with statistical precision. At the empirical controls 0.1, 0.2 and 0.3 mm, steady r was respectively 0.9424, 0.9427 and 0.9427, and winding-defect density was at most 2.4×10^{-6} . Thus this finite, single-seed parameter sweep does not show fragmentation in the empirical decay-length band; its boundary lies about sevenfold below the band’s 0.1 mm lower limit. It neither measures cortex nor proves a transition, and it leaves seed dependence and the near-boundary metastability unresolved.

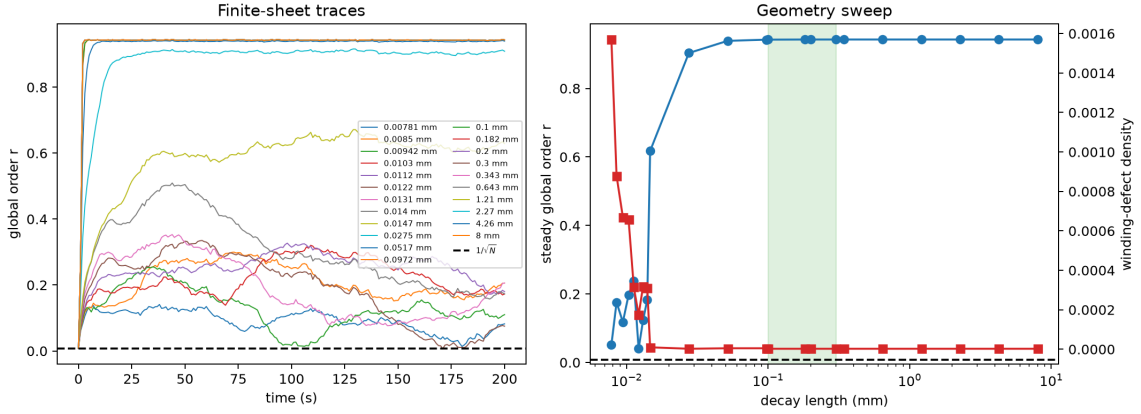


Figure 2: Finite-sheet heuristic evidence under a row-normalised spatial kernel. The left panel shows global order traces and the $1/\sqrt{N}$ finite-size floor; the right compares steady order and wrapped plaquette winding-defect density across physical decay lengths. The shaded band is 0.1–0.3 mm. This sampled simulation establishes no general transition or biological identification.

10.3 Critics

The longest-standing objection is that the fields are too weak to matter, pressed from within the field-theoretic camp by Pockett herself [?, ?]. Vöröslakos and colleagues found that applied current is shunted through scalp, delivering fields at or below 1 V/m in cortex [?]. That is a bound on *applied* rather than endogenous fields, but it constrains how much causal work a field of that magnitude can do, and is the reason we do not claim the field drives spiking.

10.4 Falsification conditions

We state the failure conditions rather than leaving them to a referee.

Epiphenomenality: if the endogenous field can be cancelled or shunted — by a conductive medium that nulls the extracellular potential without altering transmembrane currents — and neither behaviour nor report changes, the identification fails. This is the decisive experiment and it is, in principle, available.

Magnitude gap: if the spatial integration the framework requires demands field effects an order of magnitude above the $1\text{--}5 \text{ mV/mm}$ measured in cortex, the commitment fails on arithmetic. The relevant

number is not whether ephaptic coupling exists but whether its coupling strength suffices to place the system above the critical coupling K_c .

Timescale mismatch: the prediction of the main text turns on the field’s dynamics being fast relative to the astrocytic and clearance variables. If recovery from sleep inertia tracks an exponential set by the electrical timescale rather than the delayed sigmoid the slow variable predicts, the framework loses the one prediction that is its own.

Dissociation: if large-scale field coherence is preserved across states in which experiential unity is lost — or absent in states where it is retained — the identification of the global section with unified experience has no physical carrier.

11 Relationship to rival theories

Integrated Information Theory. IIT derives a scalar Φ measuring irreducibility of a system’s cause-effect structure, with the exclusion postulate selecting a single maximum- Φ complex as the locus of experience [?]. Our global section plays a structurally analogous role, but is arrived at from the opposite end: forced by a thermodynamic phase transition rather than postulated from phenomenal axioms. The unity condition is checkable from field dynamics rather than requiring the intractable computation of Φ , and exclusion falls out — gluing either succeeds or it does not. Both are vulnerable to the objection that the criterion could be satisfied by a system with no claim to experience [?]; our reply is that gluing is necessary but not sufficient, and that the topological boundary and reflexive structure are additional required conditions. IIT, by contrast, treats Φ as both necessary and sufficient.

Global Workspace Theory. GWT specifies a functional architecture without specifying what physical property of broadcasting makes the broadcast content conscious, leaving it open to the charge that it explains access consciousness while leaving phenomenal consciousness untouched [?, ?, ?]. Our framework supplies the missing substrate: the continuous macroscopic field is the medium in which integration is realized, and the $K_c = 2D$ threshold gives a physical condition for ignition rather than a functional stipulation. Where GWT is agnostic about implementation, we make a falsifiable commitment — unity requires a continuous physical medium — which predicts that a functionally equivalent discrete broadcast architecture would not be conscious. We regard this as a genuine cost of the position, not a free advantage.

Panpsychism. Consciousness is not assumed as a fundamental property of matter. It requires both a topological defect establishing a bounded “inside” and a phase transition into unity when integrated spatial coupling exceeds the noise threshold.

12 The sleep-inertia prediction: assumptions, alternatives, and falsifiability

12.1 Assumptions

Four things are assumed, and they are not small. The identification of the effective coupling K with a monotone function of extracellular geometry is a modelling assumption, not a measurement. The threshold $K_c = 2D$ uses an assumed von Mises stationary density, so the argument relies on the proved existence and continuity of the coherent branch, not the numerical threshold. The square-root cusp needs the slow variable to cross threshold at non-zero speed ($\dot{K}(t_0) \neq 0$), weaker than the adiabatic separation the whole argument assumes. And the exponent theorem proves no stability or dynamical selection — $r = 0$ remains a solution above threshold — so the time course $r(t)$ is carried by the adiabatic assumption, not a trajectory theorem. The prediction is about functional form, not a time constant: a flat foot of definite shape whose width is set by the slow variable.

12.2 Finite-ramp numerical check

A finite- N Euler–Maruyama ensemble quantifies the range over which the adiabatic reading is accurate. With identical frequencies, $D = 1$, $N = 2000$, 32 seeded replicas and $\dot{K} = v \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$, escape is defined as the first of three consecutive sampled points at $r \geq 0.2$ after K_c . The fastest leg is right-censored: only 5/32 replicas escape before $K = 2.5$. All replicas escape at the three slower

speeds, with mean delays $\Delta K = 0.1947, 0.0684$ and 0.0253 . Their log-log fit gives $\Delta K \propto v^{0.443}$, near the predicted exponent $1/2$ at this resolution. At fixed $v = 10^{-2}$ the delay increases from 0.0619 through 0.1947 to 0.2341 as N increases from 500 through 2000 to 8000 , qualitatively agreeing with the predicted $\sqrt{\log N}$ dependence; three sizes are evidence for the trend, not a scaling law.

The early-foot fit $r = A(K - K_0)^{\beta_{\text{eff}}}$ over $0.1 \leq r \leq 0.4$ gives $\beta_{\text{eff}} = 0.964, 0.898, 0.471, 0.417$ from fastest to slowest. Thus the slow ramps recover the square-root foot, while fast driving pushes the observable foot toward the linear falsifier stated below. With $D = 1.5 \text{ rad s}^{-1}$, traversing the simulated coupling window over the proposed 10^2 – 10^3 -second sleep-inertia interval corresponds to $v = 6.67 \times 10^{-4}$ – 6.67×10^{-3} in simulation units. This range straddles the crossover between the square-root-like and smeared feet, so the adiabatic shape is safe at the slow end but not established across the whole biological interval. The independently estimated (a, r) traces remain close to $I_1(a)/I_0(a)$ even where the trajectory lags the stationary branch. Post-critical RMS deviations are $0.00245, 0.00599, 0.00689, 0.00705$; using the slowest run as the finite- N floor Dev_0 , none reaches $2\text{Dev}_0 = 0.01410$. The specified critical speed v^* is therefore not bracketed by this sweep. This numerical check exhibits finite sampled dynamics only: it proves neither stability nor dynamical selection and does not discharge the adiabatic assumption.

12.3 What the alternative predicts

An account on which grogginess is the electrical realignment itself has no slow variable and no threshold. Its recovery is exponential near the fixed point: monotone from $t = 0$, fastest at the start, with no flat foot. The discriminator is the presence or absence of an initial interval of no recovery, in data already collected.

12.4 Falsification

The parameter-free form tests the von Mises stationary density and the mean-field closure jointly; it does not test $K_c = 2D$ independently of that ansatz. The prediction is falsified by: an exponential fit beating a delayed-sigmoid on early recovery; an onset preferring exponent 1 over $1/2$; a latency that does not scale with extracellular volume; a phase distribution that is not von Mises; or an (a, r) trace departing from $I_1(a)/I_0(a)$. The first two falsify with no reference to the sleep-inertia setting at all.

13 Broader philosophical context

Philosophically, the framework bridges the historical gap between type identity theory [?] and computational functionalism [?]. The subjective mental state is identified with the physical thermodynamic field, yet its qualitative nature is defined by structural resonance rather than specific biochemistry. This constrains multiple realizability: consciousness is not exclusively biological, but neither is it substrate-neutral, because the substrate is what fixes which couplings are physically available. What is proved is that fixed wiring is strictly suboptimal at matched resource, and that a finitely-sited architecture registers exactly zero coupling energy in the continuum functional; the step from either to a claim about von Neumann architectures is an informal argument.

By framing these derivations in the tradition of Russellian monism, the framework acknowledges the constitutive limits of third-person scientific explanation. The Lean 4 theorems verify the objective physical attractors, while the intrinsic “what it is like” to be that resonant field remains the domain of the first-person perspective. By locating the phase transition at a threshold with closed-form value $K_c = 2D$, it aligns with Seth’s “real problem” of consciousness [?]: rather than bridging an unbridgeable metaphysical gap, it offers a topological architecture that can explain, predict, and physically perturb the structural properties of experience.

14 Empirical test of the (a, r) collapse

This section reports the methods and results of the empirical test of the parameter-free collapse prediction $r = I_1(a)/I_0(a)$ described in the main text. The analysis was performed on a publicly available dataset of human scalp EEG recorded during propofol sedation (OpenNeuro ds005620 [?]), comprising 21 subjects with 65-channel recordings at 5 kHz.

14.1 Preprocessing

For each recording block, 20–30 seconds of data were downloaded from the public S3 bucket. An additional 3 seconds of padding on each side was downloaded and trimmed after filtering to suppress edge artifacts. Each channel was detrended and bandpass filtered with a 4th-order Butterworth filter in the 4–40 Hz band using second-order sections (`sosfiltfilt`). The instantaneous phase $\theta_j(t)$ was obtained from the Hilbert transform. Only the 62 scalp EEG channels were used; EOG and EMG channels were discarded.

14.2 (a, r) estimation

Each recording was divided into non-overlapping 100 ms bins (500 samples at 5 kHz). The estimand is the pooled spatiotemporal phase distribution over 62 sites and 100 ms, not an instantaneous 62-site spatial distribution. Pooling is required because the independent log-density estimator requires at least 100 phase samples. For each bin t :

- **Order parameter r :** $r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|$, where $N = 62$. This is exactly the Lean definition `order_parameter`.
- **Concentration a :** The slope of the log-density of the same phase distribution regressed on $\cos \theta$, as specified in the main text. Phases are centred on their circular mean, binned into 40 bins on $[-\pi, \pi]$, and log of the binned density (with a 0.5 pseudocount) is regressed on $\cos \theta$ by *unweighted* ordinary least squares. Under the von Mises model $\log \rho(\theta) = a \cos \theta - \log(2\pi I_0(a))$, so the slope is a .

The estimator must not be a maximum-likelihood one. The von Mises family is a one-parameter exponential family whose sufficient statistic is $\sum_j \cos \theta_j$ — that is, r itself — so any ML estimate of the concentration solves its score equation $I_1(\hat{a})/I_0(\hat{a}) = r$ and returns $\hat{a} = A^{-1}(r)$ identically. The pair $(\hat{a}, r)$ would then lie on $r = I_1(a)/I_0(a)$ for any input whatsoever, including phases drawn from a distribution that is not von Mises, and the collapse would be an algebraic identity rather than a test. For the same reason the regression is left unweighted: weighting bins by their counts recovers the ML score equation. Equal weight per bin is what makes the estimate depend on the shape of the distribution rather than on its first trigonometric moment.

The pseudocount biases the estimator. Against 200 synthetic replicates at this analysis’s per-bin sample size (62 channels $\times$ 500 samples = 31,000 phases, 40 bins), the bias is ≤ 0.007 for $a \leq 3$ and reaches -0.375 only at $a = 5$, where the troughs of the distribution empty out and the pseudocount dominates them; the sampling standard deviation over the same range is 0.008–0.048. All values reported below lie well inside the unbiased range.

The discriminating power of the estimator was verified directly on distributions that are not von Mises. At the same sample size, a wrapped Cauchy ($\rho = 0.6$), a top-hat arc, and a 50/50 mixture of a von Mises with a uniform give residuals $r - I_1(\hat{a})/I_0(\hat{a})$ of $+0.088$, -0.117 and $+0.075$ respectively, against a worst case of 0.0024 on genuine von Mises samples — a separation of $31.7\times$. A uniform distribution passes, correctly, since it is von Mises at $a = 0$.

14.3 Bootstrap confidence intervals

95% confidence intervals on the mean (a, r) per block were obtained by bootstrap resampling of the time bins (2000 resamples, percentile method).

14.4 Narrowband analysis

The pipeline was run separately on four narrow bands (theta 4–8 Hz, alpha 8–12 Hz, beta 13–30 Hz, gamma 30–40 Hz) and the broad band (4–40 Hz) to test whether the collapse is band-specific.

14.5 Results

Residuals are reported per 100 ms bin and then averaged, rather than computed from block-mean a and block-mean r ; because I_1/I_0 is nonlinear, the latter is not the mean residual. The estimator’s own noise floor at this sample size is 0.0024. Window sensitivity gave mean residuals -0.0261 , -0.0079 ,

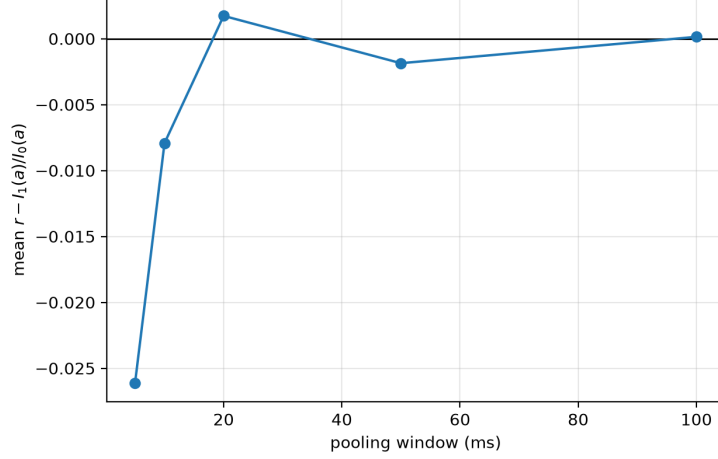


Figure 3: Sensitivity of the mean collapse residual to the spatiotemporal pooling window. This assesses the pooled estimator and does not turn 62-channel EEG into an instantaneous spatial sample.

+0.0017, -0.0019 and +0.0001 at 5, 10, 20, 50 and 100 ms respectively (Figure 3); the result is stable over 20–100 ms, but not at 5–10 ms.

- **Single subject (sub-1016, 7 blocks).** In the bipolar montage the mean residual $r - I_1(a)/I_0(a)$ lies between -0.0025 and +0.0033 in the 4–40 Hz band, between -0.0046 and +0.0046 in alpha, and between -0.0016 and +0.0043 in beta — at or below the noise floor in every block. In the common-reference montage the residual is systematically negative, between -0.041 and -0.003 (4–40 Hz) and between -0.052 and -0.036 (alpha).
- **Cross-subject (8 subjects, sed run-1).** Bipolar: mean residual -0.0002 ± 0.0002 (SEM), RMSE 0.0005, max $|r - I_1/I_0(a)| = 0.0013$. Common-reference: mean residual -0.0057 ± 0.0028 (SEM), RMSE 0.0093, max 0.0246. Group $a = 0.373$ (range 0.302–0.542) and $r = 0.173$ (range 0.145–0.222).
- **Narrowband.** Theta, alpha, beta, gamma and the broad band all collapse, with mean residuals of -0.0018, +0.0001, +0.0001, +0.0000 and +0.0000 respectively (sub-1016, sed run-1, bipolar). The collapse is not band-specific.

The gap between the two montages is the one systematic feature of the results. The common-reference residual is negative in every block and every subject, which is the signature expected of volume conduction: a shared reference adds a common-mode component that raises the resultant length r without correspondingly sharpening the shape of the phase distribution. The bipolar derivation removes it, and the residual falls by roughly an order of magnitude.

14.6 Interpretation

The (a, r) collapse to I_1/I_0 is consistent with human scalp EEG during propofol sedation, in a bipolar montage and in every frequency band tested. This is a necessary condition for the framework’s phase-distribution claim: the von Mises stationary density is compatible with the data.

The strength of that confirmation is limited by the range of a the data occupy, and we state the limit rather than leave it to a referee. Across the eight subjects a spans only 0.302–0.542. Over so short an arc the Bessel ratio is nearly straight: $I_1(a)/I_0(a)$ departs from the naive linear form $a/2$ by at most 0.0094 across that interval, and from $\tanh(a/2)$ by at most 0.0030, while the per-bin residual scatter is 0.02–0.03. The data therefore confirm that r and a stand in the relation the theory requires, but they do not discriminate I_1/I_0 from a straight line, and no claim of having done so should be read into them. Discriminating the curve from its tangent needs recordings that reach $a \gtrsim 1.5$, where the two separate by more than 0.17; the sleep-inertia protocol of the main text is attractive partly because a recovery sweeping through threshold traverses exactly that range.

The result does not test the empirical commitment that the endogenous electromagnetic field is the coupling medium, nor the stipulation that the global section is experience. The two sleep-inertia

predictions — the temporal shape (delayed sigmoid vs. exponential) and the threshold condition ($a/r > 2$)
— remain untested on human data.